

Properties of the arithmetic operations

The five laws that make algebra possible, written for letters instead of numbers.

LESSON 60

1 × 40 MIN

MATEMATIKA 7, §26

S8 2.1

LESSON CLOCK

10'

12'

12'

6'

The five laws	10 min
What is not a law	12 min
Using them	12 min
Homework	6 min

LEARNING OBJECTIVES

- State the commutative, associative and distributive laws.
- Use them to reorder and regroup a calculation.
- Recognise that subtraction and division are not commutative.
- Use the laws to simplify an expression quickly.

TEXTBOOK

National	pp. 160–163
Cambridge	Stage 8, pp. 12–18
Workbook	Exercise 2.1

01

Explanation

The five laws

LAW	STATEMENT	SAYS
commutative (+)	$a + b = b + a$	order does not matter
commutative (×)	$ab = ba$	order does not matter
associative (+)	$(a + b) + c = a + (b + c)$	grouping does not matter
associative (×)	$(ab)c = a(bc)$	grouping does not matter
distributive	$a(b + c) = ab + ac$	the two operations connect

Two more, about the special numbers 0 and 1:

$a + 0 = a$ $a \times 1 = a$ $a \times 0 = 0$

THE DISTRIBUTIVE LAW IS THE ONLY ONE THAT MIXES + AND ×

Everything in algebra that turns a sum into a product, or a product into a sum, is that one law — expanding brackets, factorising, and collecting like terms alike.

What is not a law

CLAIM	TRUE?	COUNTER-EXAMPLE
$a - b = b - a$	no	$5 - 3 \neq 3 - 5$
$a \div b = b \div a$	no	$6 \div 2 \neq 2 \div 6$
$(a - b) - c = a - (b - c)$	no	$(9 - 4) - 2 \neq 9 - (4 - 2)$
$a + (b \times c) = (a + b)(a + c)$	no	$2 + 3 \times 4 \neq 5 \times 6$

ONE COUNTER-EXAMPLE IS A COMPLETE DISPROOF

A rule that fails once is not a rule. Testing a suspicious claim with small numbers is the fastest check in all of algebra.

Using them

The laws are used to make a calculation easier, not to make it longer.

CALCULATION	LAW USED	EASY ROUTE
$25 \times 17 \times 4$	commutative and associative	$(25 \times 4) \times 17 = 1700$
7×98	distributive	$7(100 - 2) = 686$
$36 + 47 + 64$	commutative	$(36 + 64) + 47 = 147$
$4a + 7b - a + 2b$	commutative	$3a + 9b$

LOOK FOR PAIRS THAT MAKE ROUND NUMBERS

25 with 4, 36 with 64, 98 with 100. Spotting them before starting turns a hard calculation into an easy one.

02 Worked examples**EXAMPLE 1** Compute $25 \times 17 \times 4$ in the easiest way.

① Reorder: $25 \times 4 \times 17$.

Commutative.

② Regroup: $(25 \times 4) \times 17$.

Associative.

③ $= 100 \times 17$

④ $= 1700$

ANSWER 1700 **EXAMPLE 2** Compute 7×98 without a calculator.

① $98 = 100 - 2$

② $7(100 - 2)$

Distributive.

③ $= 700 - 14$

④ $= 686$

ANSWER 686

EXAMPLE 3 Show that subtraction is not associative.

- ① Test $(9 - 4) - 2$ and $9 - (4 - 2)$.
- ② $5 - 2 = 3$
- ③ $9 - 2 = 7$
- ④ $3 \neq 7$ — one counter-example is enough.

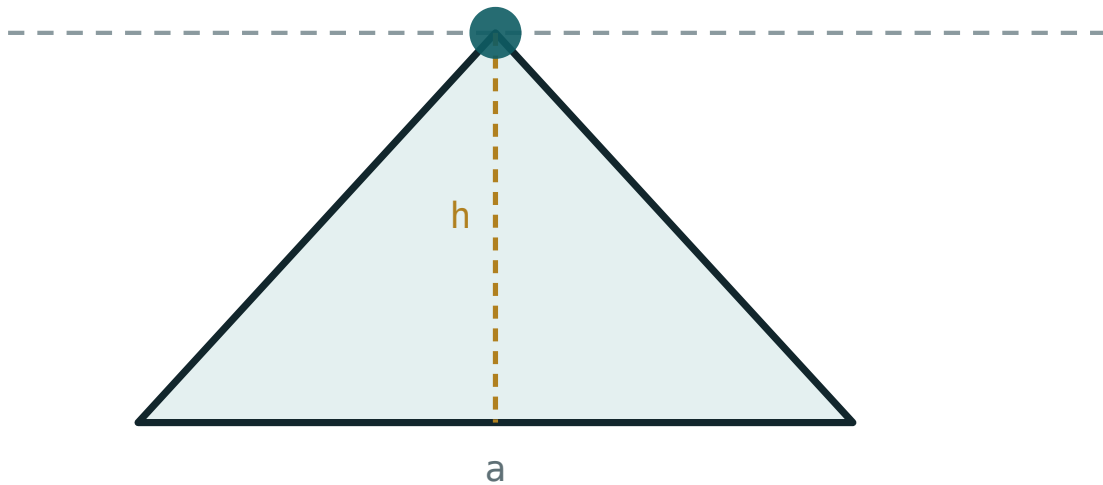
ANSWER

Disproved

03 Interactive model

Ask for $25 \times 17 \times 4$ with no calculator; the class that reorders finishes in five seconds and the rest do not.

The distributive law as an area

base a **220**height h **120**area $\frac{1}{2}ah$ **13200**perimeter **545.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Commutative law	O'rin almashtirish qonuni	Переместительный закон
Associative law	Guruhlash qonuni	Сочетательный закон
Distributive law	Taqsimot qonuni	Распределительный закон
Identity element	Neytral element	Нейтральный элемент
To reorder	Tartibini o'zgartirish	Переставить
To regroup	Guruhlarni o'zgartirish	Перегруппировать
Not commutative	Kommutativ emas	Не коммутативно
Counter-example	Qarama-qarshi misol	Контрпример

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$a + b = b + a$ is:

associative

commutative

distributive

not a law

$(ab)c = a(bc)$ is:

associative

commutative

distributive

not a law

$a(b + c) = ab + ac$ is:

associative

commutative

distributive

not a law

Is subtraction commutative?

yes

no

sometimes

only for positives

The easiest route to $25 \times 17 \times 4$:

left to right

pair 25 with 4

pair 17 with 4

a calculator

One counter-example:

weakens a rule

disproves it

proves it

means nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Name the law: $a + b = b + a$

ANSWER Commutative

2. Name the law: $(ab)c = a(bc)$

ANSWER Associative

3. Name the law: $a(b + c) = ab + ac$

ANSWER Distributive

4. Is $a - b = b - a$?

ANSWER No

5. Is $a \div b = b \div a$?

ANSWER No

6. $a + 0$

ANSWER a

7. $a \times 1$

ANSWER a

Medium · the standard question

7 PROBLEMS

1. $25 \times 17 \times 4$

ANSWER 1700

2. 7×98

ANSWER 686

3. $36 + 47 + 64$

ANSWER 147

4. $8 \times 125 \times 6$

ANSWER 6000

5. 12×101

ANSWER 1212

6. Simplify $4a + 7b - a + 2b$

ANSWER $3a + 9b$

7. Simplify $5x - 2y + 3y - x$

ANSWER $4x + y$ **Hard · stretch and reason**

7 PROBLEMS

1. $99 \times 99 + 99$

ANSWER 9900

2. $37 \times 25 \times 4$

ANSWER 3700

3. $16 \times 47 + 16 \times 53$

ANSWER 1600

4. Show that $(a - b) - c \neq a - (b - c)$

ANSWER $(9-4)-2 = 3$, $9-(4-2) = 7$

5. Simplify $2(3a - b) - 3(a - 2b)$

ANSWER $3a + 4b$

6. 101×99

ANSWER 9999

7. Why is the distributive law special?

ANSWER It is the only one linking + and $\times$

07 Homework

Homework — 5 tasks

Look for the easy pairing before computing anything.

1. Compute $4 \times 37 \times 25$ using the laws.
2. Compute 9×102 using the distributive law.
3. Compute $58 + 79 + 42$ by reordering.
4. Show with an example that division is not commutative.
5. Simplify $6a - 2b + 3b - 4a$.